LAB 6
LIGHT INTENSITY & DISTANCE

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OBJECTIVE AND EXPERIMENT

Determine how detected light intensity from a "point" source varies with distance, and check whether the inverse-square law $I = P/(4\pi r^2)$ holds.

- Principle: Light from a point source spreads over a sphere of area $4\pi r^2$, so intensity falls as $1/r^2$.
- Setup: 1.2 m optical track, 12 V / 10 W G4 halogen source, PASCO light sensor (resolution $1\ \mu\mathrm{Lux}$).
- Method: Record steady-state intensity at $d = 0, 10, 20, 30, 40, 50\ \mathrm{cm}$; record background with light off.
- Analysis: Subtract background, fit $I(r)$, and verify that $4\pi E r^2$ is constant across distance.
- Expected: $I \propto 1/r^2$; the recovered "power" $P = 4\pi b$ should be consistent within uncertainty.

METHOD

- Run 1 (light ON): hold sensor at each distance until plateau, record ~100–200 samples.
- Run 2 (light OFF): record ~1670 background samples.
- For each plateau: compute mean and standard deviation
- Subtract the background mean from each plateau mean
- Combine type A (statistical) + type B (sensor scale, rectangular PDF) in quadrature:

$$\sigma_E = \sqrt{s_{platue}^2 + s_{bg}^2 + \left({\sigma_{res}}/{\sqrt{3}}\right)^2}$$

- Fit $I(r) = b/(r + r_0)^2 \Rightarrow$ recover $P = 4\pi b$
- Check χ^2/ν ,then cross-check that $4\pi \cdot E \cdot r_{eff}^2$ is constant across distance.

DATA AND MEASUREMENTS

Plateau means (Run 1) and standard deviations at each distance, with background subtracted (mean of Run 2 = 0.0969%).

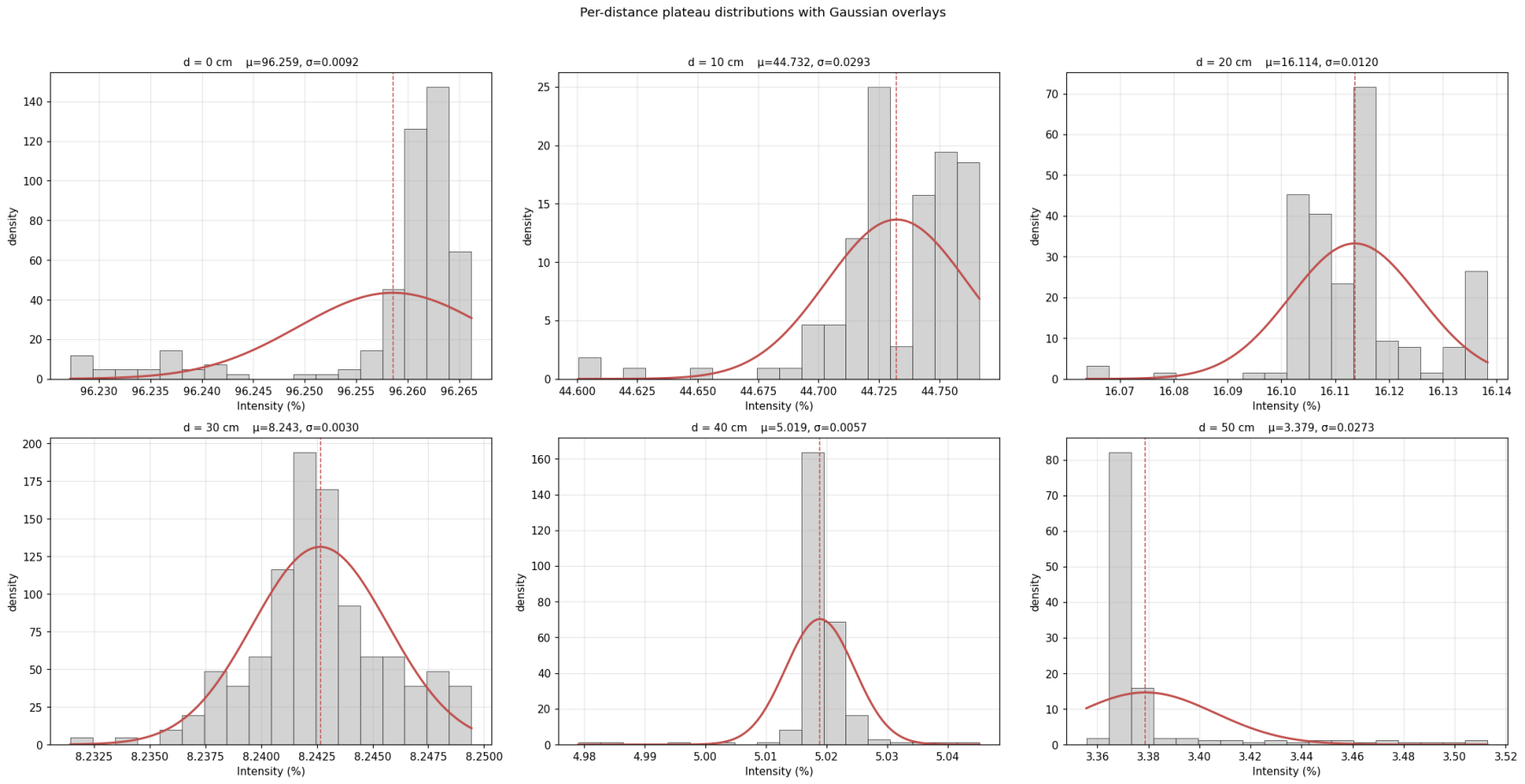
d (cm)	N	$mean\ I$ (%)	std I (%)	$mean - bg$ (%)
0	194	96.2586	0.00918	96.1616
10	117	44.7323	0.02925	44.6354
20	155	16.1137	0.01200	16.0167
30	206	8.2426	0.00304	8.1457
40	198	5.0189	0.00568	4.9220
50	201	3.3788	0.02726	3.2819

Background (Run 2): $N = 1669$, mean $= 0.0969\%$, $s = 0.0012\%$ (Gaussian).

Distance uncertainty: $\sigma_r = 5\ \mathrm{mm}/\sqrt{3} \approx 2.9\ \mathrm{mm}$ (rectangular PDF; ruler resolution + endpoint).

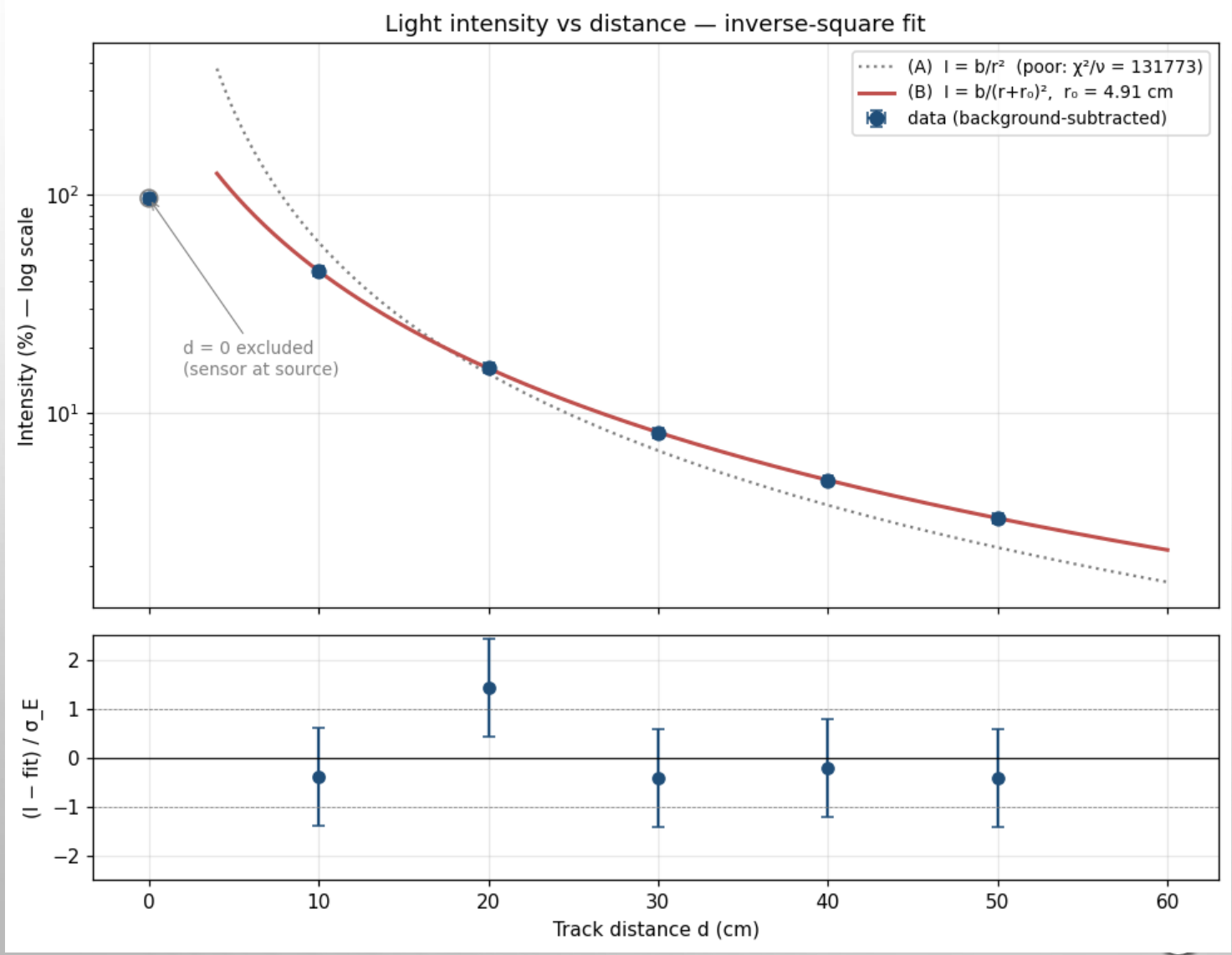
Sensor resolution σ_{res}: 1×10^{-6} Lux (rectangular PDF) — negligible vs plateau std.

PER-DISTANCE DISTRIBUTIONS



- Each plateau yields one mean and one standard deviation.
- Histograms peak around the mean and span a finite range.
- Approximately Gaussian shape — normal PDF justified for σ_E
- Plateau std is the dominant (type A) contribution to σ_E
- Background and $1\ \mu\mathrm{Lux}$ sensor resolution add negligibly in quadrature.
- Standard deviations: $0.003 - 0.029\%$ across the six distances.

INVERSE-SQUARE FIT



Fit form

- $I(r) = b/(r + r_0)^2 \Rightarrow P = 4\pi b$
- r_0 = offset between marked $d = 0$ and the actual filament inside the bulb.
- $d = 0$ excluded (sensor essentially at source — point-source approx fails).
- Reduced χ^2:

$$\chi_\nu^2 = \frac{1}{\nu}\Sigma_i \frac{(I_i - \hat{I}_i)^2}{\sigma_{E,i}^2}$$

Best-fit results:

$$b = 0.9932 \pm 0.0008 \ \% \cdot m^2$$

$$r_0 = 4.91 \pm 0.01 \text{ cm}$$

$$\frac{\chi^2}{\nu} = 0.85 \ (\nu = 3)$$

WHY INCLUDE r_0?

- True distance form the source to sensor is $r = d + r_0$, where r_0 is the offset to be determined

It's the value that minimizes the weighted sum of squared residuals i.e., the fit that best balances all five data points ($d = 10, 20, 30, 40, 50$ cm) given their individual uncertainties.

The uncertainty ± 0.01 cm comes from the covariance matrix that the fitter computes at the minimum quantifying the precision of the recovered parameter.

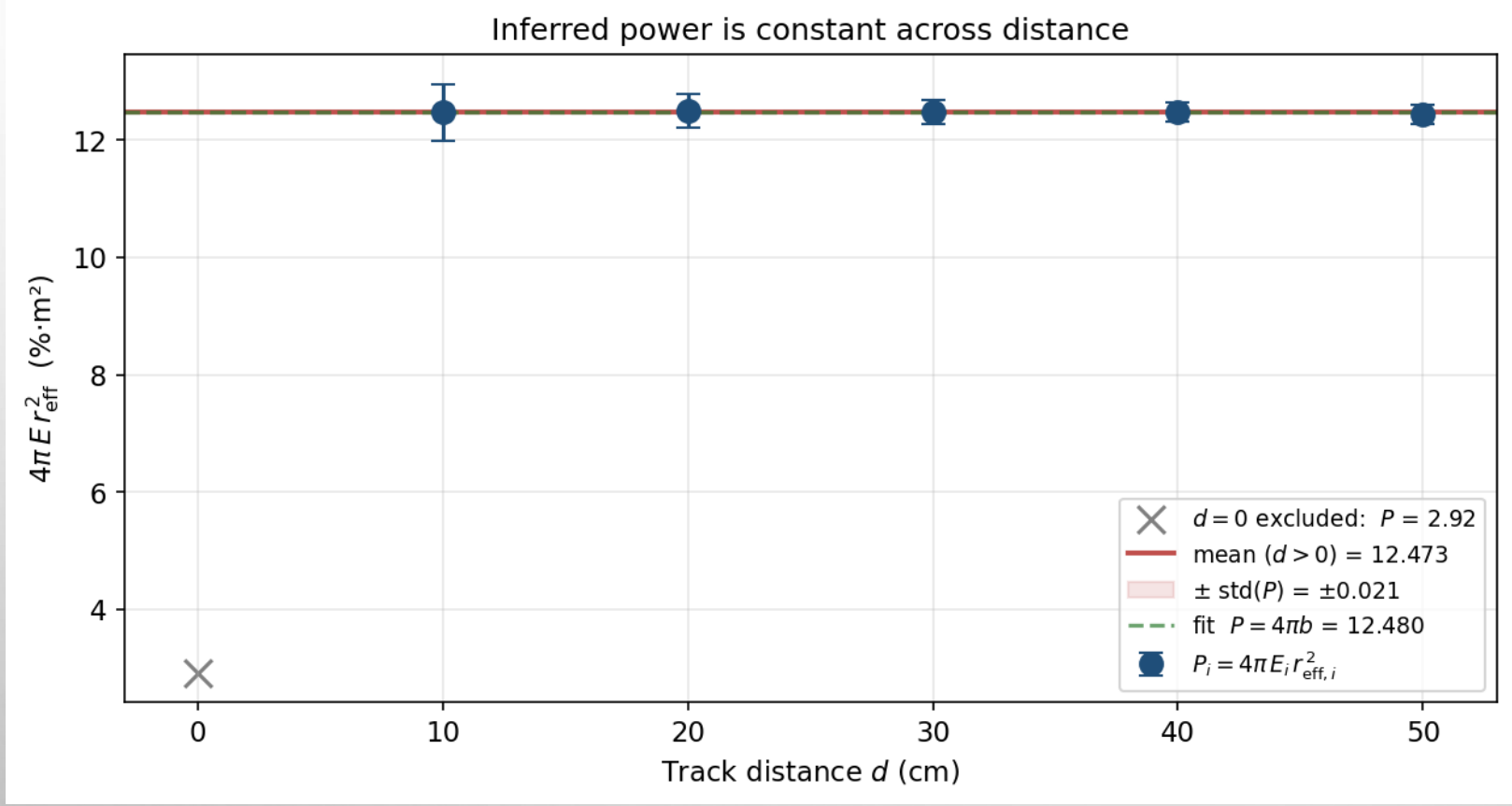
Evidence from data

- Fitting $I = b/r^2$ gave $\chi^2/\nu \approx 131{,}773$ overestimating intensity at small distances and underestimating at large distances
- Adding r_0 as a free parameter dropped $\chi^2/\nu = 0.85$

Found using `scipy.optimize.curve_fit()` with $I(r) = b/(r + r_0)^2$ and minimizing $\chi^2_\nu = \frac{1}{\nu}\Sigma_i \frac{(I_i - \hat{I}_i)^2}{\sigma^2_{E,i}}$.

Varies both b and r_0 simultaneously until it finds the smallest χ^2

IS THE INFERRED POWER CONSTANT?



Per-point power

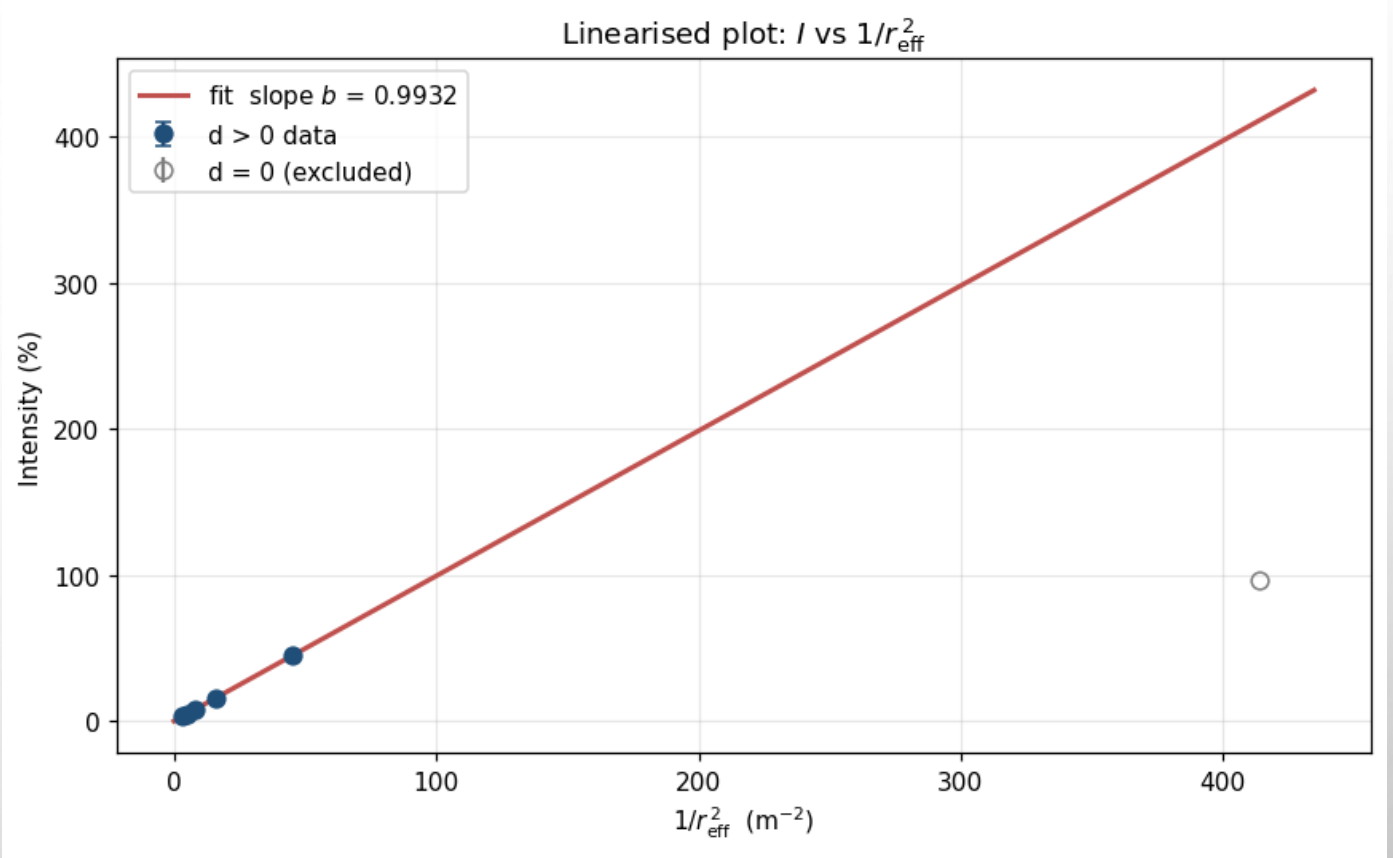
- Compute $P_i = 4\pi E_i r_{eff} i^2$ for each distance.
- Propagate σ_r and σ_E in quadrature:

$$\sigma_P = \sqrt{(8\pi E r \sigma_r)^2 + (4\pi r^2 \sigma_E)^2}$$

- All five $P_i(d > 0)$ agree to $\sim 0.2\%$
- $\mu_P = 12.473, \quad \sigma_P = 0.021$
- $1/r^2$ behavior is confirmed.

LINEARIZATION

If $I = b/r_{\text{eff}}^2$ holds, plotting I against $1/r_{\text{eff}}^2$ should be linear through the origin with slope b



COMPARISON AND SIGNIFICANCE

t-test: cross-check vs fit

$$t = \frac{|P_{exp} - P_{fit}|}{\sqrt{\sigma_{exp}^2 + \sigma_{fit}^2}}$$

- $P_{exp} = 12.473 \pm 0.009$ (mean of 5 distances)
- $P_{fit} = 12.480 \pm 0.010$ (least-squares fit)
- $t = 0.56 \Rightarrow$ fully consistent (Gaussian PDF, $t < 1$).
- Recall: $t = 1 \rightarrow 68\%$ probability of difference; $t = 2 \rightarrow 95\%$; $t = 3 \rightarrow 99.7\%$

Sanity check vs source rating

- Source: G4 halogen, 12 V, 10 W (electrical input).
- Halogen luminous efficacy $\approx 10 - 15\ \mathrm{lm/W} \Rightarrow \approx 100 - 150$ lumens.
- Sensor reads % of full scale (not lux) — direct numeric comparison needs the % → lux calibration.

CONCLUSIONS

$$P_{exp} = 12.480 \pm 0.010\ \% \cdot \mathrm{m}^2$$
$$r_0 = 4.92 \pm 0.01\ \mathrm{cm}$$

Gaussian PDF, 68% confidence level

Key findings

- Inverse-square law confirmed: $\chi^2/\nu = 0.85$, $t = 0.56$, residual pulls within $\pm 2\sigma$
- Recovered source-position offset $r_0 = 4.91\ \mathrm{cm}$ (filament behind front of bulb housing).
- $P_i = 4\pi \cdot E \cdot r_{eff}^2$ is constant across $d = 10$ to $50\ \mathrm{cm}$ to $\sim 0.2\%$
- Order of magnitude consistent with manufacturer rating (12 V / 10 W halogen ≈ 100 lm).
- Limitations: sensor in % units (no lux calibration); single trial; finite source size breaks point-source approximation near $d = 0$.